65. *De Hebdomadibus* of Boethius II.24, cited in C. H. Kahn, *Anaximander and the Origins of Greek Cosmology* (New York: Columbia University Press, 1960), pp. 84–85.

66. Some examples of important arguments where Einstein has strongly supported a thesis that he later changed his mind about: 1. The expansion of the universe (first ridiculed, then accepted); 2. The existence of gravitational waves (first taken as obvious, then rejected, then accepted again); 3. The equations of relativity do not admit solutions without matter (a long-defended thesis that was abandoned—rightly so); 4. Nothing exists beyond the horizon of Schwarzschild (wrong, though perhaps he never came to realize this); 5. The equations of the gravitational field cannot be general-covariant (asserted in the work with Grossmann in 1912; three years later, Einstein argued the opposite); 6. The importance of the cosmological constant (first affirmed, then denied—having been right the first time.) . . .

8. DYNAMICS AS RELATION

67. The general form of a mechanical theory that describes the development of a system *in time* is given by a phase space and a Hamiltonian *H*. Evolution is described by the orbits generated by *H*, parametrized by the time *t*. The general form of a mechanical theory that describes the evolutions of *variable with respect to each other* is instead given by a phase space and a constraint *C*. The relations between the variables are given by the orbits generated by *C* in the subspace *C=0*. The parametrization of these orbits has no physical meaning. A detailed technical discussion can be found in chapter 3 of Carlo Rovelli, *Quantum Gravity* (Cambridge, UK: Cambridge University Press, 2004). For a concise technical account, see Carlo Rovelli, "Forget Time," *Foundations of Physics* 41 (2011): 1475–90, https://arxiv.org/abs/0903.3832.

68. An accessible account of loop quantum gravity can be found in Rovelli, *Reality Is Not What It Seems* (op. cit.).

69. Bryce S. DeWitt, "Quantum Theory of Gravity. I. The Canonical Theory," *Physical Review* 160 (1967): 1112–48.

70. J. A. Wheeler, "Hermann Weyl and the Unity of Knowledge," *American Scientist* 74 (1986): 366–75.

71. J. Butterfield and C. J. Isham, "On the Emergence of Time in Quantum Gravity," in *The Arguments of Time*, ed. J. Butterfield (Oxford: Oxford University Press, 1999), pp. 111–68 (http://philsci-archive.pitt.edu/1914/1/EmergTimeQG=9901024.pdf); Zeh, *Die Physik der Zeitrichtung*; Craig Callender and Nick Huggett, eds., *Physics Meets Philosophy at the Planck Scale* (Cambridge, UK: Cambridge University Press, 2001); Sean Carroll, *From Eternity to Here: The Quest for the Ultimate Theory of Time* (New York: Dutton, 2010).

72. The general form of a quantum theory that describes the evolution of a system *in time* is given by a Hilbert space and the Hamiltonian operator *H*. The evolution is described by Schrödinger's equation $i\hbar\partial_t\Psi = H\Psi$. The probability of measuring a pure state Ψ a time *t* after having measured a state Ψ' is given by the transition amplitude $\langle\Psi \mid \exp[-iHt/\hbar] \mid \Psi'\rangle$. The general form of a quantum theory that describes the evolution of the variables *with respect to one another* is given by a Hilbert space and Wheeler-DeWitt equation $C\Psi = 0$. The probability of measuring the state Ψ after having measured the state Ψ' is determined by the amplitude $\langle\Psi \mid \int dt \exp[iCt/\hbar]|\Psi'\rangle$. A detailed technical discussion can be found in chapter 5 of Rovelli, *Quantum Gravity* (op. cit.). For a concise technical version, see Rovelli, "Forget Time" (op. cit.)

73. Bryce S. DeWitt, *Sopra un raggio di luce* (Rome: Di Renzo, 2005).

74. There are three: they define the Hilbert space of the theory where the elementary operators are defined, whose eigenstates describe the quanta of space and the probability of transitions between these.